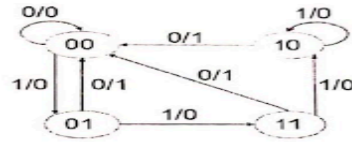


1.C04

Given state diagram as follows, get the sequential circuit using JK flip-flop:



Present State		Input	Next State		Output	Flip Flop Inputs			
A	B	X	A ⁺	B ⁺	Y	JA	KA	JB	KB
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	X
0	1	0	0	0	1	0	X	X	1
0	1	1	1	1	0	1	X	X	0
1	0	0	0	0	1	X	1	0	X
1	0	1	1	0	0	X	0	0	X
1	1	0	0	0	1	X	1	X	1
1	1	1	1	0	0	X	0	X	1

A \ Bx	Bx'	B'x	Bx	Bx'
A'	0	1	1	0
A	X	X	X	X

$JA = Bx$

A \ Bx	Bx'	B'x	Bx	Bx'
A'	X	X	X	X
A	1	0	0	1

$KA = X'$

A \ Bx	Bx'	B'x	Bx	Bx'
A'	0	1	X	X
A	0	0	X	X

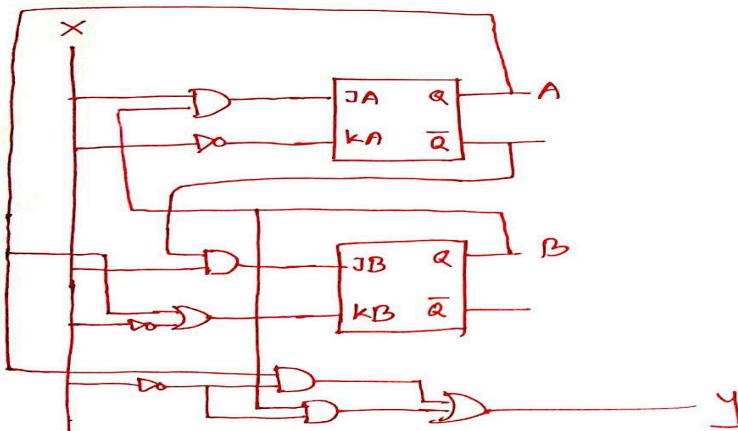
$JB = Ax$

A \ Bx	Bx'	B'x	Bx	Bx'
A'	X	X	0	1
A	X	X	1	1

$KB = X' + A$

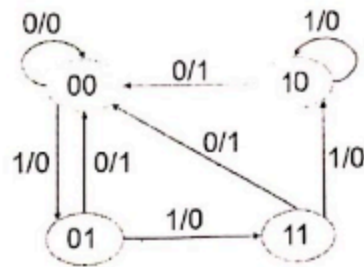
A \ Bx	Bx'	B'x	Bx	Bx'
A'	0	0	0	1
A	1	0	0	1

$Y = Ax' + Bx'$

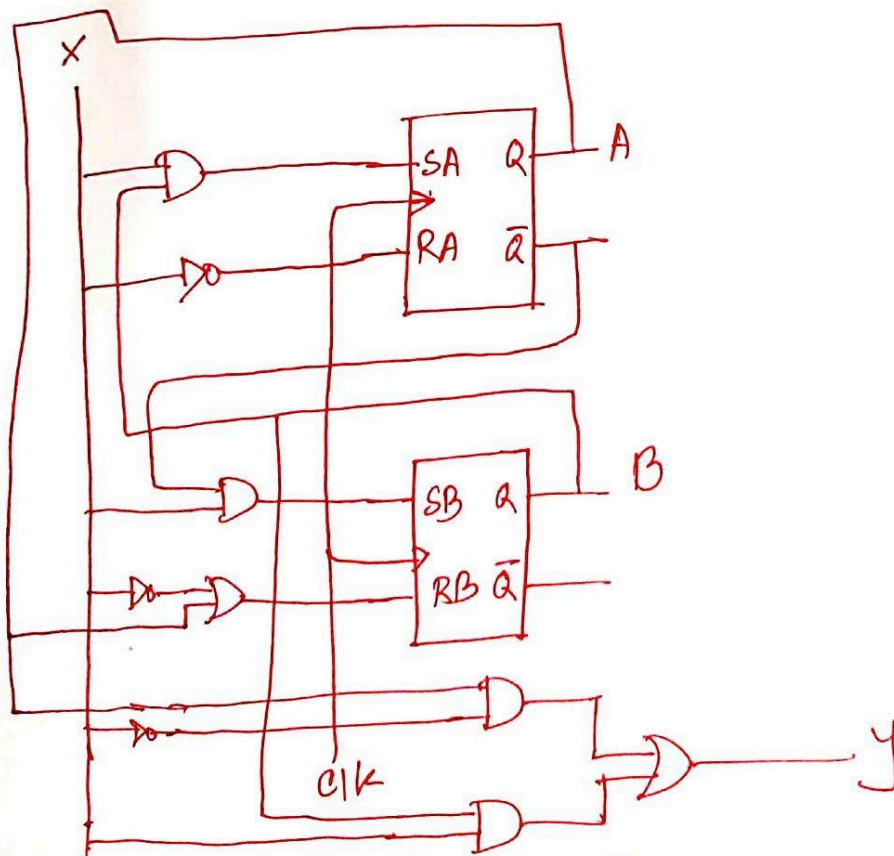
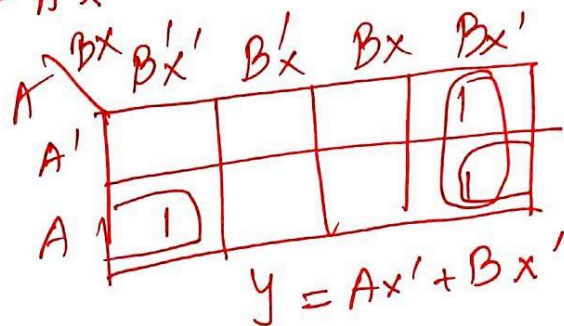
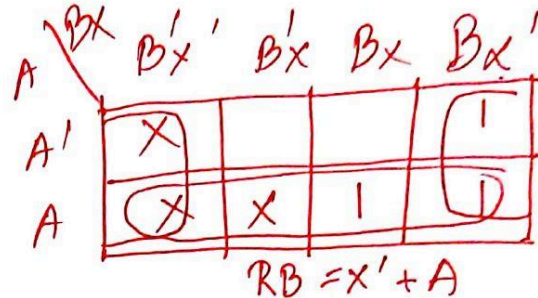
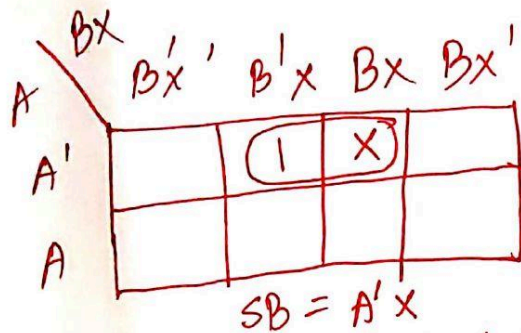
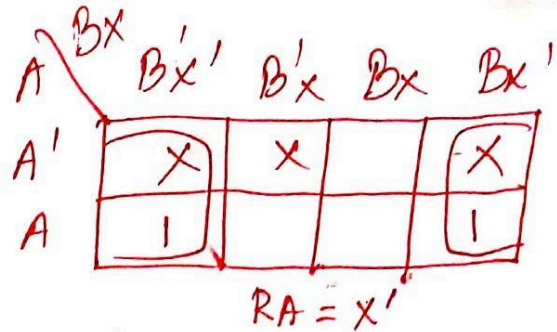
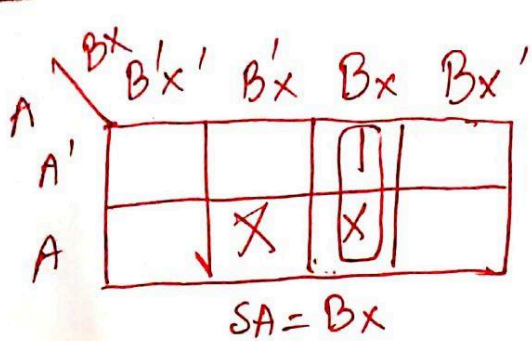


1.CO4

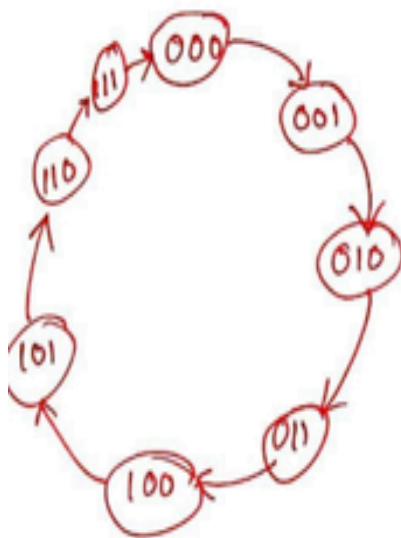
Given state diagram as follows, get the sequential circuit using SR flip-flop:



Present State		Input	Next State		Output	Flip Flop Inputs			
A	B		A ⁺	B ⁺		SA	RA	SB	RB
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	0
0	1	0	0	0	1	0	X	0	1
0	1	1	1	1	0	1	0	X	0
1	0	0	0	0	1	0	1	0	X
1	0	1	1	0	0	X	0	0	X
1	1	0	0	0	1	0	1	0	1
1	1	1	1	0	0	X	0	0	1



1.CO4 Implement the following counter using JK FF: 0→1→2→3→4→5→6→7→0



Present State			Next State			Flip Flop Inputs					
A ₂	A ₁	A ₀	A ₂ ⁺	A ₁ ⁺	A ₀ ⁺	J _{A2}	K _{A2}	J _{A1}	K _{A1}	J _{A0}	K _{A0}
0	0	0	0	0	1	0	x	0	x	1	x
0	0	1	0	1	0	0	x	1	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	0	1	x	0	0	x	1	x
1	0	1	1	1	0	x	0	1	x	x	1
1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	0	0	0	x	1	x	1	x	1

		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'			1	
	A_2	x	x	x	x

$$JA_2 = A_1 A_0$$

		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'		1	x	x
	A_2		1	x	x

$$JA_1 = A_0$$

		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'	1	x	x	1
	A_2	1	x	x	1

$$JA_0 = 1$$

		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'	x	x	1	x
	A_2			1	

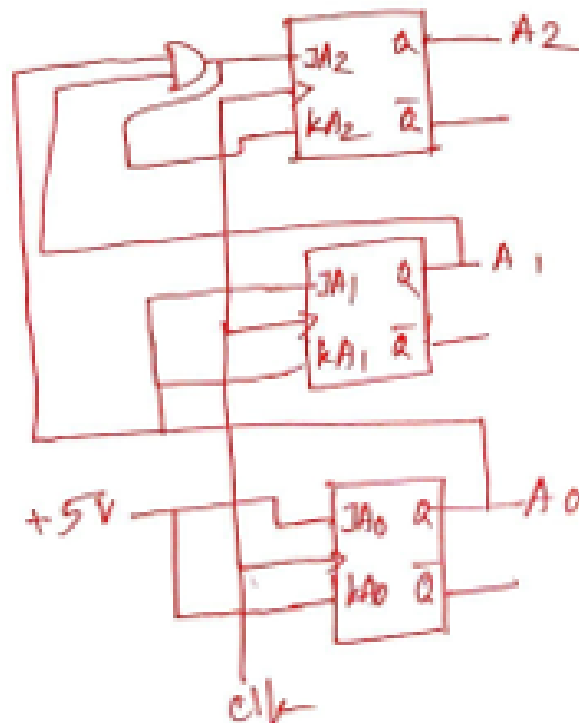
$$KA_2 = A_1 A_0$$

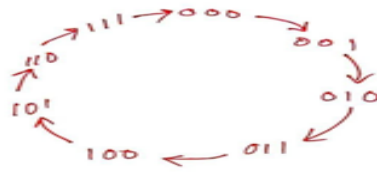
		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'	x	x	1	
	A_2	x	x	1	

$$KA_1 = A_0$$

		$A_1 A_0$			
		$A_1' A_0'$	$A_1' A_0$	$A_1 A_0$	$A_1 A_0'$
A_2	A_2'	x	1	1	x
	A_2	x	1	1	x

$$KA_0 = 1$$





Present State			Next State			Flip Flop Inputs					
A_2	A_1	A_0	A_2^+	A_1^+	A_0^+	SA_2	RA_2	SA_1	RA_1	SA_0	RA_0
0	0	0	0	0	1	0	X	0	X	1	0
0	0	1	0	1	0	0	X	1	0	0	1
0	1	0	0	1	1	0	X	X	0	1	0
0	1	1	1	0	0	1	0	0	1	0	1
1	0	0	1	0	1	X	0	0	X	1	0
1	0	1	1	1	0	X	0	1	0	0	1
1	1	0	1	1	1	X	0	X	0	1	0
1	1	1	0	0	0	0	1	0	1	0	1

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'	0	1	1	0
A_2	X	X	X	X

$SA_2 = A_2'A_1A_0$

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'	X	X		X
A_2			1	

$RA_2 = A_2A_1A_0$

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'		1	0	X
A_2		1		X

$SA_1 = A_1'A_0$

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'	X			1
A_2	X			1

$RA_1 = A_1A_0$

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'	1			1
A_2	1			1

$SA_0 = A_0'$

A_2	A_1A_0	$A_1'A_0'$	$A_1'A_0$	A_1A_0'
A_2'		1	1	
A_2		1	1	

$RA_0 = A_0$

